



Google Cloud Memorystore

Fully Managed In-Memory Data Store for Modern Applications

Fast · Scalable · Reliable

Redis Compatible

Valkey Support

Sub-millisecond Latency

99.99% SLA

A Technical Overview for Cloud Engineers

🔗 What is Memorystore?

Fully managed in-memory data store service

🗄️ In-Memory Database Service

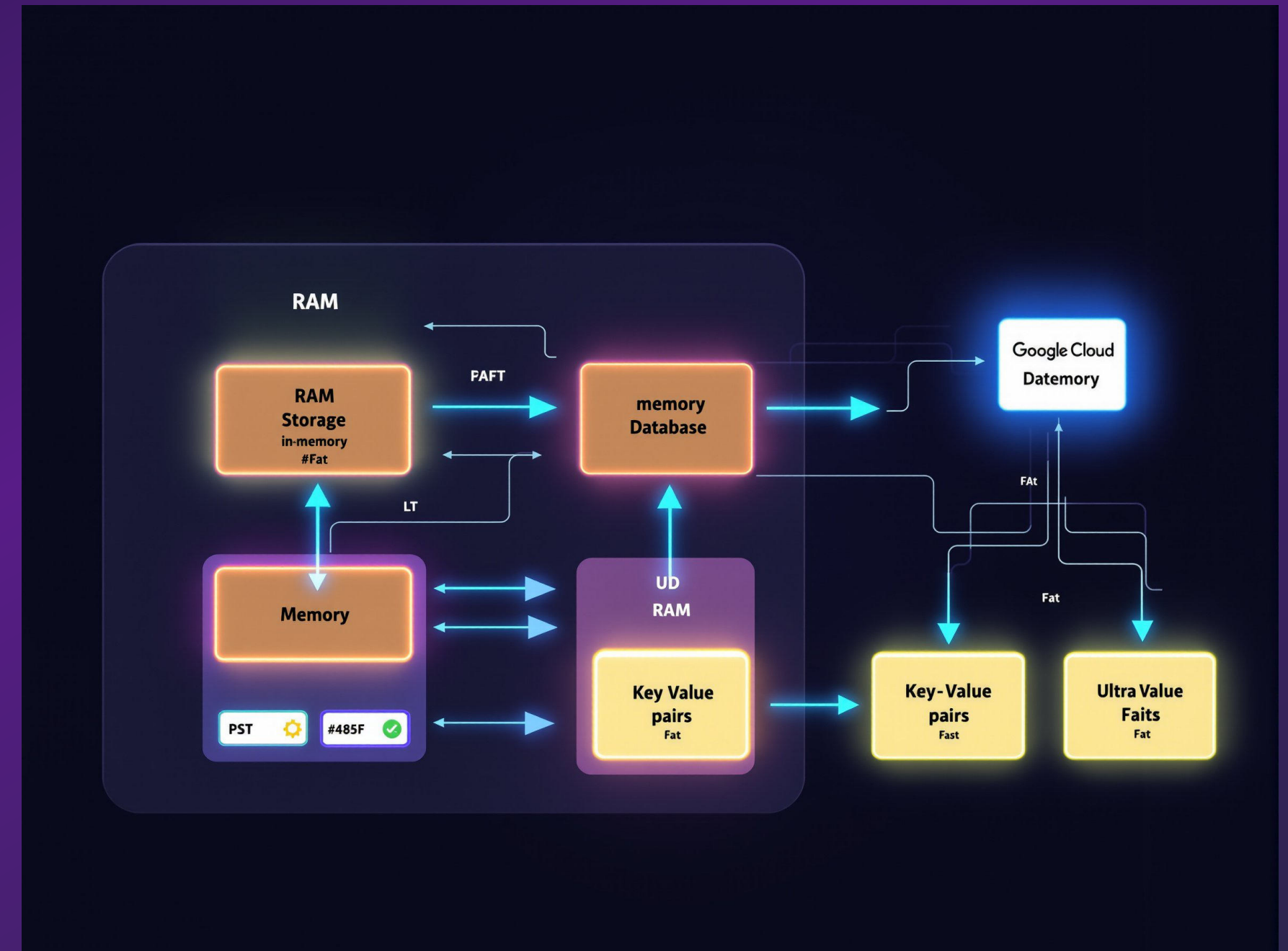
🔑 Key-Value Store Architecture

⚡ Sub-millisecond Latency

☁️ Google Managed Infrastructure

60x

More throughput than standalone instances



Protocol Compatibility



Redis Protocol

Redis 7.2 and earlier
API Compatible

Zero code changes required

- Advanced data structures
- Pub/Sub messaging
- Transactions



Memcached Protocol

100% compatible
Full Protocol Support

Drop-in replacement

- Simple key-value operations
- String data types
- High-speed caching

Cross-Platform Client Support



Python

redis-py, python-memcached



Java

Jedis, Lettuce,
spymemcached



Node.js

node-redis, memcached



Go

go-redis, gomemcache



.NET

StackExchange.Redis



PHP/Ruby

Various clients

Seamless Integration with Existing Redis and Memcached Applications

Fully Managed by Google

 **80-90%**
Reduction in Infrastructure Management Time
Focus on application development, not infrastructure operations



Automated Provisioning

Automatic instance creation and configuration



Built-in Replication

Automatic data replication across availability zones



Automated Failover

Automatic failover with minimal downtime



Seamless Patching

Security updates without user intervention



Integrated Monitoring

Built-in Cloud Monitoring with dashboards



Automated Backups

Scheduled backups and recovery automation

Zero Infrastructure Management, Maximum Development Focus

Ultra-High Performance



Sub-millisecond

Standard Operation Latency



Microseconds

Optimized Latency (Valkey/Redis)



60x Throughput

vs. Standalone Redis



250 Nodes

Maximum Cluster Size



Terabytes

Data Capacity

Zero-downtime scaling maintains performance



OpenX Case Study

150+B

Ad Requests Daily

Sub-millisecond latency for billions of daily ops.

OpenX

In-Memory Operations Deliver Unmatched Speed

↗ Scalability & High Availability

↗ Scalability Features

↑ **Vertical Scaling**
Increase instance size for more capacity.

↔ **Horizontal Scaling**
Add nodes/shards up to 250 nodes.

✓ **Zero-Downtime Scaling**
Scale without service interruption.

↻ **Automatic Rebalancing**
Engine-driven slot migration.

🛡 High Availability

↻ **Automatic Failover**
Detects failures and promotes replicas.

🌐 **Multi-Zone Replication**
Replicas across availability zones.

📖 **Cross-Region Support**
Geographic distribution for Valkey & Redis.

💾 **Persistent Storage**
RDB snapshots and AOF logging.

Service Level Agreements

🏆
99.99%

Redis Cluster & Valkey

Enterprise-grade availability

✓
99.9%

Redis Standard Tier

Production-ready reliability

⚠
No SLA

Redis Basic Tier

Development and testing

Scale from Gigabytes to Terabytes with Consistent Performance

Typical Use Cases



Caching

- Database query caching
- Content caching (HTML, API)
- Application-level caching

Reduces database load by 80%

Latency: seconds to milliseconds



Session Management

- User authentication sessions
- Shopping cart persistence
- Cross-server session sharing

Maintain cart state across servers

Sub-millisecond session lookup



Counters & Analytics

- Real-time page view counters
- API rate limiting
- Gaming leaderboards

Real-time leaderboards for millions of players

Atomic increments for real-time analytics



Pub/Sub Messaging

- Real-time notifications
- Live chat applications
- IoT device commands

Broadcast messages to thousands of users

Low-latency message delivery

✧ Memorystore for Valkey

The Open-Source Evolution

Launched March 2024

<> Open-Source Fork of Redis

BSD 3-clause license - truly open source



Linux Foundation Backed

Contributors: Google, Amazon, Snap, and more



Redis 7.2 API Compatible

Zero code changes for migration



Ongoing Innovation

Active development with community contributions

Valkey Journey



March 2024

Valkey OSS Launch



August 2024

Google Cloud Preview



October 2024

Valkey 8.0 Support



April 2, 2025

General Availability



April 22, 2025

Vector Search GA

The future of open-source in-memory databases

Open Source · Community-Driven · Future-Ready

Valkey Features & Benefits



99.99% SLA

Enterprise-Grade



2x Performance

with Valkey 8.0



250 Nodes

Maximum Scale



Vector Search

AI/ML Ready



Valkey 8.0 Enhancements

- Up to 2x QPS improvement
- Asynchronous I/O threading
- Automatic failover for empty shards
- Improved memory efficiency
- Slot migration recovery

Preview Available



Vector Search (GA)

- Ultra-low latency similarity search
- Billions of vectors supported
- Approximate Nearest Neighbor (ANN)
- Exact Nearest Neighbor (KNN)
- Hybrid search with filters

RAG, Recommendations, Semantic Search



High Availability

- Zero-downtime scaling
- Multi-zone replica distribution
- Cross-region replication
- Automated failover
- Multi-VPC access (20 VPCs)

Maximum uptime and reliability

Latest Innovation with Open-Source Foundation

Redis Cluster Architecture

Distributed Data Architecture



Instance

Top-level resource with configuration



Shards

Data partitions distributing keyspace



Nodes

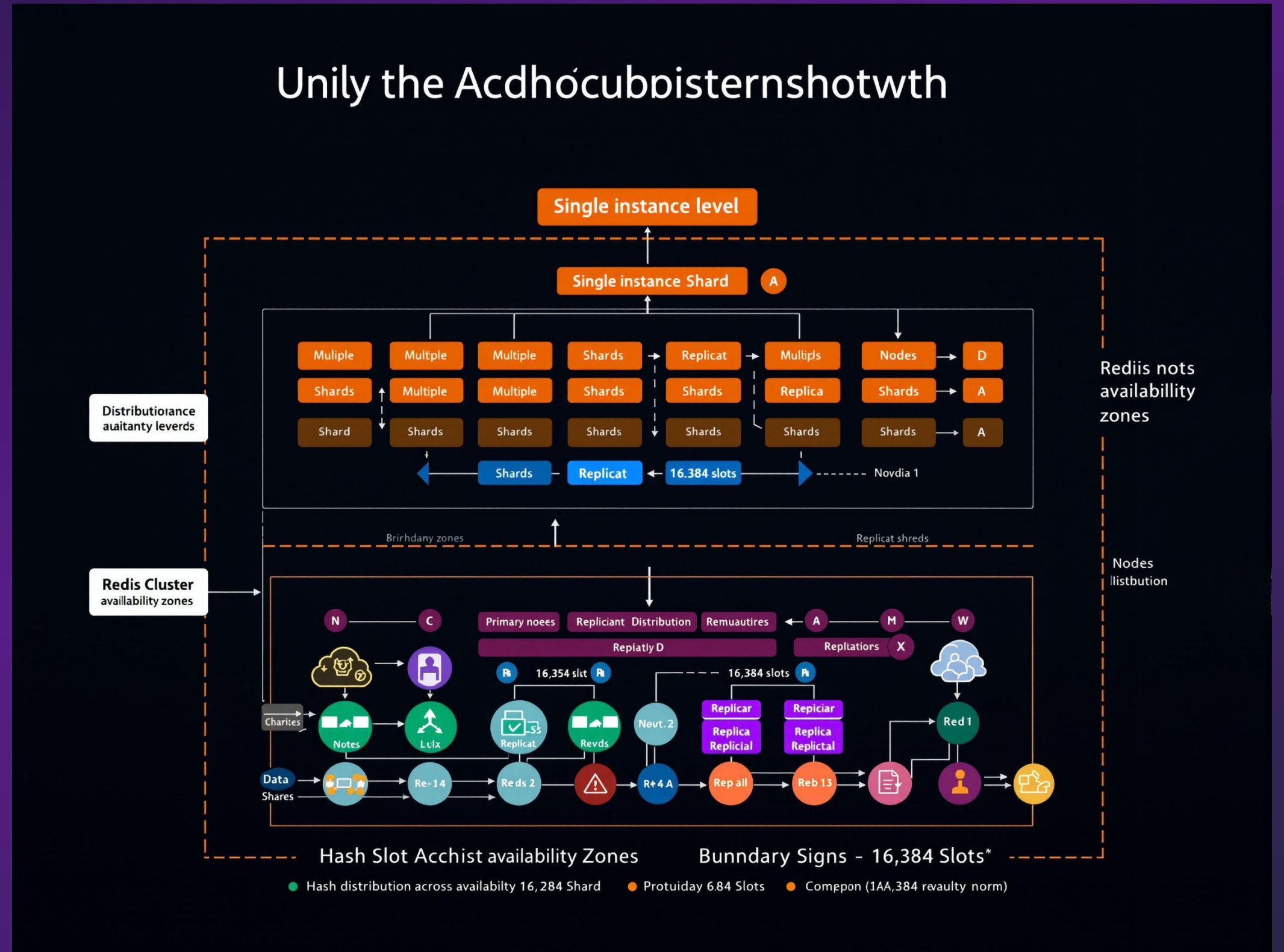
Individual Redis processes (primary + replicas)

16,384

Hash Slots

Keyspace divided and distributed across shards

Each shard manages a portion of hash slots



Horizontal Scaling through Intelligent Data Sharding

↗ Redis Cluster: Scale & Availability



250 Nodes

Maximum Cluster Size



60x

More Throughput than Standalone



99.99%

Availability SLA



Zero-Downtime Scaling



Engine-Driven Migration

Built into Redis engine, no external agents



Automatic Rebalancing

Smart data redistribution during scaling



OOM Prevention

Intelligent memory management prevents errors



High Availability



Multi-Zone Replication

Automatic replica distribution across zones



Automatic Failover

Instant replica promotion on primary failure



Cross-Region Support

Geographic distribution for disaster recovery



Memorystore Variants Comparison

Feature	Memcached	Redis Standalone	Redis Cluster	Valkey
Data Structures	String only	Advanced (Lists, Sets, Hashes)	Advanced	Advanced
Persistence	None	RDB + AOF	RDB + AOF	RDB + AOF
High Availability	No	Yes (Standard Tier)	Yes	Yes
SLA	None	99.9%	99.99%	99.99%
Max Data Size	Memory-limited	Up to 300GB	Terabytes	Terabytes
Throughput	High	Moderate	60x Redis standalone	60x + 2x with v8.0
Horizontal Scaling	Manual	No	Yes (250 nodes)	Yes (250 nodes)
Vector Search	No	No	No	Yes (GA)
License	BSD (Open)	Source-available	Source-available	BSD (Open)
Cost	Low	Medium	High	High
Best For	Simple caching	General-purpose	Massive scale	Future-proof + AI/ML

Choose Based on Your Scale, Availability, and Feature Requirements

Decision Guide: Which Variant to Choose?

What is your primary requirement?

Simple, fast caching only?

No persistence, basic key-value, budget-conscious

Need advanced features?

Complex data structures, persistence, pub/sub

< 100GB

& <100K ops/sec

> 100GB

or >100K ops/sec



Memcached

Best for simple, fast caching without advanced features



Redis Standalone

Small to medium workloads, 99.9% SLA

Need AI/ML features?



Valkey

Vector search + massive scale



Redis Cluster

Massive scale without AI

Follow the Path to Your Optimal Solution

Access from Google Cloud

VPC Networking & Connectivity

 Private IP Access Only - No Public Internet Exposure



Direct Peering

Legacy mode for automatic VPC peering

Older instances



Private Service Access

Recommended for flexibility & Shared VPC

Recommended



Private Service Connect

Modern approach for Redis Cluster

Latest

Compute Services Integration



Compute Engine

Direct VPC access from VMs



Google Kubernetes Engine

VPC-native clusters with pod access



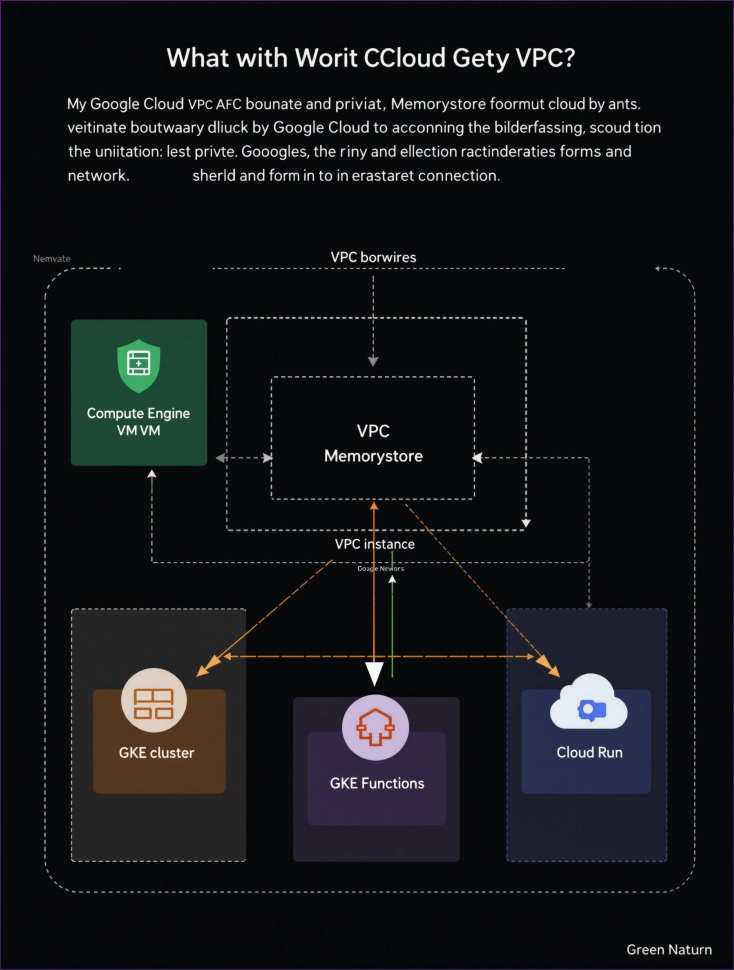
Cloud Functions

Serverless access via VPC connector



Cloud Run

Containerized app access via connector



Secure, Private Connectivity within Google Cloud Ecosystem

Security & Getting Started

Enterprise-Grade Security

- **Network Security**
- VPC network isolation
 - Private IPs & Firewall rules

- **Authentication**
- IAM integration
 - Redis AUTH passwords

- **Encryption**
- In-transit (TLS)
 - At-rest with CMEK support

- **Monitoring**
- Cloud Audit Logs
 - Access & compliance reporting

Why Choose Memystore?

**Fully Managed**

Reduced infrastructure management

**High Performance**

Sub-millisecond latency

**Highly Available**

99.99% SLA for Cluster variants

**Scalable**

Scale from GBs to TBs, no downtime

**Secure**

Enterprise-grade security built-in

**Future-Ready**

Vector search, Valkey 8.0 support

Ready to Get Started?

Explore Documentation & Quickstarts

Activate Your \$300 Free Trial Credit

Thank you! Questions?